# Worksheet 5.2 — Related Rates

Grade 12 Calculus and Vectors (MCV4U) · Unit 5: Applications of Derivatives

## The Big Ideas

Write an equation relating the quantities, differentiate both sides with respect to $t$, then substitute the instant and solve for the unknown rate.

- Identify known/unknown rates.
- Relate the quantities; differentiate w.r.t. $t$ (chain rule).
- Substitute the instant's values; solve.

## The Big Picture

### Rates linked in time

In a **related rates** problem several quantities change together over time, and their rates are connected by an equation.

### Differentiate implicitly

Differentiate the relating equation with respect to time $t$, so each variable's rate appears through the chain rule.

### Substitute last

Plug in the known values only *after* differentiating, then solve for the unknown rate.

## Worked Examples

### Example 1 — Expanding circle

$\frac{dr}{dt} = 2$ cm/s. Find $\frac{dA}{dt}$ at $r = 5$.

***Solution.*** **Step 1 (write the relating equation):** $A = \pi r^2$.

**Step 2 (differentiate both sides with respect to $t$, using the chain rule on $r^2$):** $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

**Step 3 (substitute $r = 5$ and $\frac{dr}{dt} = 2$):** $\frac{dA}{dt} = 2\pi(5)(2)$.

**Conclusion:** $\frac{dA}{dt} = 20\pi \approx 62.8$ cm$^2$/s.

## Example 2 — Sliding ladder

A 13 m ladder's base slides out at 2 m/s. Find the top's rate when the base is 5 m out.

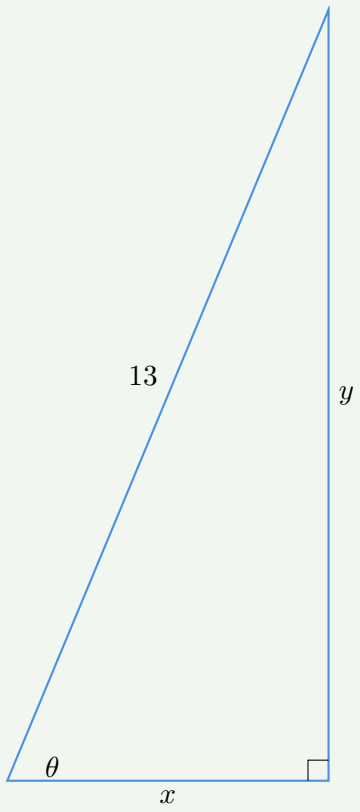
***Solution.*** **Step 1 (write the relating equation, Pythagorean theorem):** $x^2 + y^2 = 169$.

**Step 2 (find $y$ at this instant):** $x = 5 \Rightarrow y = \sqrt{169 - 25} = \sqrt{144} = 12$.

**Step 3 (differentiate both sides with respect to $t$):** $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0$.

**Step 4 (substitute known values and solve for $\frac{dy}{dt}$):** $2(5)(2) + 2(12)\dfrac{dy}{dt} = 0 \Rightarrow \dfrac{dy}{dt} = -\dfrac{20}{24}$.

**Conclusion:** $\dfrac{dy}{dt} = -\dfrac{5}{6}$ m/s (negative — the top is sliding down):




## Example 3 — Inflating balloon

$\dfrac{dV}{dt} = 100$ cm$^3$/s. Find $\dfrac{dr}{dt}$ at $r = 5$.

***Solution.*** **Step 1 (write the relating equation):** $V = \frac{4}{3}\pi r^3$.

**Step 2 (differentiate both sides with respect to $t$):** $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$.

**Step 3 (substitute $r = 5$ and $\frac{dV}{dt} = 100$):** $100 = 4\pi(25)\dfrac{dr}{dt} = 100\pi\dfrac{dr}{dt}$.

**Conclusion:** $\dfrac{dr}{dt} = \dfrac{1}{\pi}$ cm/s.

### Example 4 — Growing square

A square's side grows at 3 cm/s. Find $\frac{dA}{dt}$ at $s = 10$.

***Solution.*** **Step 1 (write the relating equation):** $A = s^2$.

**Step 2 (differentiate both sides with respect to $t$):** $\frac{dA}{dt} = 2s\frac{ds}{dt}$.

**Step 3 (substitute $s = 10$ and $\frac{ds}{dt} = 3$):** $2(10)(3)$.

**Conclusion:** $\frac{dA}{dt} = 60$ cm$^2$/s.

### Example 5 — Two cars

Car A north at 60, car B east at 80 km/h from a point. Find the rate the distance grows after 1 h.

***Solution.*** **Step 1 (write the relating equation):** if $a, b$ are the cars' distances travelled and $D$ the distance between them, $D^2 = a^2 + b^2$.

**Step 2 (find the values after 1 h):** $a = 60,\ b = 80 \Rightarrow D = \sqrt{60^2 + 80^2} = \sqrt{10000} = 100$.

**Step 3 (differentiate both sides with respect to $t$):** $2D\frac{dD}{dt} = 2a\frac{da}{dt} + 2b\frac{db}{dt}$.

**Step 4 (substitute and solve for $\frac{dD}{dt}$):** $\frac{dD}{dt} = \frac{60(60) + 80(80)}{100} = \frac{3600 + 6400}{100}$.

**Conclusion:** $\frac{dD}{dt} = 100$ km/h.

### Example 6 — Expanding circle

$\frac{dr}{dt} = 3$ cm/s. Find $\frac{dA}{dt}$ at $r = 4$.

***Solution.*** **Step 1 (differentiate $A = \pi r^2$ with respect to $t$):** $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$.

**Step 2 (substitute $r = 4$, $\frac{dr}{dt} = 3$):** $2\pi(4)(3)$.

**Conclusion:** $\frac{dA}{dt} = 24\pi$ cm$^2$/s.

### Example 7 — Growing square

Side grows at 2 cm/s. Find $\frac{dA}{dt}$ at $s = 6$.

***Solution.*** **Step 1 (differentiate $A = s^2$ with respect to $t$):** $\frac{dA}{dt} = 2s\frac{ds}{dt}$.

**Step 2 (substitute $s = 6$, $\frac{ds}{dt} = 2$):** $2(6)(2)$.

**Conclusion:** $\frac{dA}{dt} = 24$ cm$^2$/s.

## Example 8 — Inflating balloon

$\frac{dV}{dt} = 36\pi$ cm$^3$/s. Find $\frac{dr}{dt}$ at $r = 3$.

***Solution.*** **Step 1 (differentiate $V = \frac{4}{3}\pi r^3$ with respect to $t$):** $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.

**Step 2 (substitute $r = 3$, $\frac{dV}{dt} = 36\pi$):** $36\pi = 4\pi(9)\frac{dr}{dt} = 36\pi\frac{dr}{dt}$.

**Conclusion:** $\frac{dr}{dt} = 1$ cm/s.

## Example 9 — Sliding ladder

A 10 m ladder's base slides out at 1 m/s. Find the top's rate at base = 6 m.

***Solution.*** **Step 1 (write the relating equation and find $y$):** $x^2 + y^2 = 100$; at $x = 6$, $y = \sqrt{100 - 36} = \sqrt{64} = 8$.

**Step 2 (differentiate with respect to $t$):** $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$.

**Step 3 (substitute and solve):** $2(6)(1) + 2(8)\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{12}{16}$.

**Conclusion:** $\frac{dy}{dt} = -\frac{3}{4}$ m/s.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

A circle's radius grows at 5 cm/s. Find $\frac{dA}{dt}$ at $r = 2$.

### Question 2

A square's side grows at 4 cm/s. Find $\frac{dA}{dt}$ at $s = 5$.

### Question 3

A balloon fills at $\frac{dV}{dt} = 48\pi$ cmş/s. Find $\frac{dr}{dt}$ at $r = 2$.

### Question 4

A 25 m ladder's base slides out at 3 m/s. Find the top's rate at base = 7 m.

### Question 5

A circle's radius grows at 1 cm/s. Find $\frac{dA}{dt}$ at $r = 10$.

## Question 6

What's the first step after writing the relating equation?

## Question 7

A square's area grows; side grows at 2 cm/s. Find $\frac{dA}{dt}$ at $s = 8$.

## Question 8

A balloon fills at $\frac{dV}{dt} = 4\pi$ cmş/s. Find $\frac{dr}{dt}$ at $r = 1$.

## Question 9

Two cars: N at 30, E at 40 km/h. Rate the distance grows after 1 h?

## Question 10

A 5 m ladder's base slides out at 1 m/s. Find the top's rate at base = 3 m.

## Question 11

What does a negative rate of change indicate?

## Question 12

A circle's area grows at $20\pi$ cmš/s. Find $\frac{dr}{dt}$ at $r = 5$ (use $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$).

## Question 13 — Challenge

Water fills a cylinder (radius 3) so the volume grows at $18\pi$ cmş/s. How fast is the height rising? ($V = \pi r^2 h$)

# Question Recap & Answer Key

## Question Recap

1. A circle's radius grows at 5 cm/s. Find $\frac{dA}{dt}$ at $r = 2$.
2. A square's side grows at 4 cm/s. Find $\frac{dA}{dt}$ at $s = 5$.
3. A balloon fills at $\frac{dV}{dt} = 48\pi$ cmş/s. Find $\frac{dr}{dt}$ at $r = 2$.
4. A 25 m ladder's base slides out at 3 m/s. Find the top's rate at base = 7 m.
5. A circle's radius grows at 1 cm/s. Find $\frac{dA}{dt}$ at $r = 10$.
6. What's the first step after writing the relating equation?
7. A square's area grows; side grows at 2 cm/s. Find $\frac{dA}{dt}$ at $s = 8$.
8. A balloon fills at $\frac{dV}{dt} = 4\pi$ cmş/s. Find $\frac{dr}{dt}$ at $r = 1$.
9. Two cars: N at 30, E at 40 km/h. Rate the distance grows after 1 h?
10. A 5 m ladder's base slides out at 1 m/s. Find the top's rate at base = 3 m.
11. What does a negative rate of change indicate?
12. A circle's area grows at $20\pi$ cmš/s. Find $\frac{dr}{dt}$ at $r = 5$ (use $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$).
13. Water fills a cylinder (radius 3) so the volume grows at $18\pi$ cmş/s. How fast is the height rising? ($V = \pi r^2 h$)

## ✓ Answer Key

1. $20\pi$ cmš/s
2. 40 cmš/s
3. $48\pi = 4\pi(4)\frac{dr}{dt} \Rightarrow 3$ cm/s
4. $y = 24$; $-\frac{7(3)}{24} = -\frac{7}{8}$ m/s
5. $20\pi$ cmš/s
6. differentiate w.r.t. $t$
7. 32 cmš/s
8. $4\pi = 4\pi\frac{dr}{dt} \Rightarrow 1$ cm/s
9. $D = 50$; $\frac{30^2+40^2}{50} = 50$ km/h
10. $y = 4$; $-\frac{3(1)}{4} = -\frac{3}{4}$ m/s
11. the quantity is decreasing
12. $20\pi = 2\pi(5)\frac{dr}{dt} \Rightarrow 2$ cm/s
13. $18\pi = 9\pi\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = 2$ cm/s